# Obed Junias

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## Professional Summary

Software Engineer with experience building scalable ML and NLP systems, distributed pipelines, and cloud-native infrastructure. Strong background in Python, SQL, production ML, and Kubernetes, with industry experience at Oracle and HPE. Interested in data/ML roles involving large-scale data processing and cloud.

## Industry Experience

**Senior Member of Technical Staff** at Oracle Corporation — Aug 2021 - Aug 2024

- Designed and implemented scalable automation frameworks in Python, TestNG, and BATS, **reducing manual validation effort by 95%** across Oracle REST Data Services and SQLcl.
- Built a Python-based migration framework to automate Oracle APEX application transfers, improving deployment reliability and supporting **200+ internal users**.
- **Led validation and release readiness** for a new Oracle Cloud DB Tools feature, coordinating a **15-engineer** effort and delivering production readiness in 2 months.

**Machine Learning Engineer Intern** at Hewlett Packard Enterprise — Feb 2021 - Jul 2021

- Built intelligent automation pipelines using Groovy and WorkFusion OCR to process unstructured documents, **improving processing speed by 40% and reducing errors by 80%.**
- Integrated OCR outputs into enterprise automation workflows and delivered a production-ready proof of concept to cross-functional stakeholders.

## Relevant Projects

- **Adapter-Based Multi-Agent LLM Serving System**
  - Built a **custom multi-agent LLM serving system** executing 14 task-specialized agents as dynamic **LoRA adapters** over a shared Llama-3B backbone, and **benchmarked against vLLM-style serving**.
  - Implemented **dynamic adapter loading, eviction, and scheduling** under strict **5–15 GB GPU memory budgets**, streaming adapters from CPU to GPU at runtime.
  - Evaluated high-entropy agentic workloads (up to **140 adapter switches per run**) across multiple schedulers, benchmarking **latency, throughput, GPU utilization, and switching overhead**.
  - Demonstrated that **LoRA adapters load in less than 1s** versus **9-22s full-model loads**, yielding **20-90x lower cumulative switching overhead** and enabling execution where full models are infeasible.
  - **Tech:** vLLM, PyTorch, HuggingFace Transformers, PEFT/LoRA, CUDA, Python
- **End-to-End Text-to-Comic Generation Platform**
  - Built a **production-grade, cloud-native AI system** that converts natural-language narratives into comic strips using **LLMs and diffusion models**, with a **React frontend** and **Node.js backend**.
  - Designed a **modular, asynchronous pipeline** deployed on **GKE**, using **Pub/Sub** for orchestration, **Cloud Storage** for persistence, and **Redis** for hot-path caching.
  - Implemented **OAuth 2.1-based authentication** and per-request isolation, supporting **100+ users** and sustaining **20–30 concurrent generation jobs**.
  - Reduced median end-to-end latency by **35–45%** through caching, pipeline parallelism, and reuse of intermediate artifacts under concurrent load.
  - **Tech:** React, Node.js, Python, GKE, Pub/Sub, Redis, Cloud Storage, OAuth 2.1, GPT, Stable Diffusion
- **GradCompass: Multi-Agent LLM Application Assistant**
  - Built a **production-grade multi-agent LLM system** for personalized graduate-application guidance, document review, and mock interview simulation.
  - Designed an **8-agent architecture** with role-specialized agents and a centralized orchestration layer.
  - Integrated a **Retrieval-Augmented Generation (RAG)** pipeline to ground recommendations in curated program and admissions data.
  - Implemented agent coordination and response synthesis for coherent outputs across multi-step workflows.
  - **Tech:** Python, LLM APIs, RAG, Vector Databases, Multi-Agent Orchestration

- **Lost in Plot: Contrastive Learning-Based Movie Retrieval**
  - Built a semantic retrieval system for **"tip-of-the-tongue" movie search**, handling vague and incomplete natural-language queries, and constructed a **100K+ example synthetic dataset** for training and evaluation.
  - Fine-tuned a **BERT-based dual encoder** using a **multi-task contrastive loss** to align user descriptions with structured movie metadata.
  - Outperformed a **GPT-4 few-shot baseline** on **Recall@K and MRR**, with strong retrieval quality.
  - **Tech:** PyTorch, BERT, Contrastive Learning, Information Retrieval

## Skills

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|---|---|
| **Languages** | Python, SQL, Hive, Java, C++ |
| **ML Techniques** | Regression, Naive Bayes, Random Forests, Gradient Boosting, SVMs, Neural Networks |
| **AI/ML Frameworks** | PyTorch, TensorFlow, Transformers, LangChain, CUDA |
| **Distributed Data** | Spark, Hadoop, MapReduce, in-memory processing, distributed file systems |
| **Tools & Platforms** | Git, Docker, Kubernetes, Redis, MongoDB, Selenium |
| **Cloud Technologies** | AWS, GCP, Oracle Cloud (OCI) |

## Education

**M.S. in Computer Science**, **University of Colorado Boulder** — Aug 2024 – May 2026
*Courses:* NLP, Deep Natural Language Understanding, Datacenter-Scale Computing — (GPA: 4.0/4.0)

**B.E. in Computer Science**, BMS College of Engineering — Aug 2017 – Aug 2021
*Relevant Courses:* Algorithms, Databases, Operating Systems, Machine Learning

## Research Experience

**Responsible LLM Evaluation for Mental Health Applications**
*Peer-reviewed (IEEE-EMBS 2025)*

- Built large-scale **evaluation pipelines** to measure demographic bias and performance of LLMs (LLaMA, GPT-4) for mental health classification.
- Implemented **in-context prompting** and **fairness-aware losses** to mitigate bias in low-resource settings.
- Developed diagnostic metrics to quantify subgroup performance, consistency, and bias across model variants.

**Commonsense Reasoning with Logical Entailment Trees**

- Designed structured benchmarks and evaluation pipelines to assess multi-step commonsense reasoning in LLMs.
- Analyzed failure modes of chain-of-thought and n-shot prompting under compositional reasoning tasks.

## Publications

- **Junias, Obed***, et al. *Assessing Algorithmic Bias in Language-Based Depression Detection: A Comparison of DNN and LLM Approaches.* IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2025. (Peer-reviewed, Presented). https://ieeexplore.ieee.org/abstract/document/11269509
- **Junias, Obed**, and Maria Leonor Pacheco. *LOGICAL-COMMONSENSEQA: A Benchmark for Logical Commonsense Reasoning.* arXiv preprint arXiv:2601.16504, 2026. https://arxiv.org/abs/2601.16504

## Awards and Recognition

- **NSF–EMBS–Google Young Professional NextGen Scholar**, *IEEE EMBS BHI Conference*, 2025

## TECHNICAL INTERESTS

Distributed Systems • ML Systems • Scalable LLM Applications • Multi-Agent Architectures